

Assignment-1

→ MVC Design Pattern Summary

• Overview

→ The model - View - Controller (MVC) design pattern divides an application into three interconnected components:

- Model: Manages the data and business logic.
- View: Handles the presentation layer and user Interface.
- Controller: Processes user input and interacts with Model and View.

User → Controller → Model → View → User

→ Variants of MVC

1. Model - View - Presenter (MVP):

- Presenter replaces Controller.
- View more passive; Presenter updates the View.

When to use: Desktop or form-based applications where clear separation between UI and logic is needed.

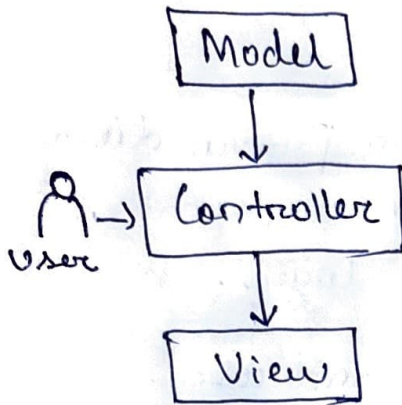
2. Model - View - ViewModel (MVVM):

- ViewModel binds View and Model using data-binding (Common in WPF, Angular).
- View Model acts as a data converter and mediator.

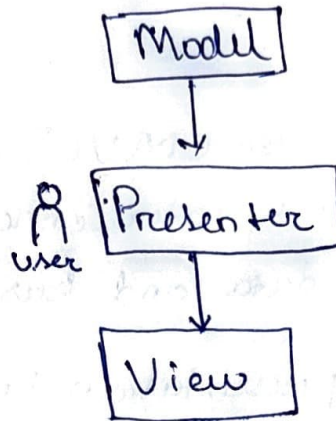
When to use:

Modern frontend frameworks (Angular, React with State Management) where data-binding is key.

MVC



MVP



MVVM

